Diameter Variability of Hypercubes*

Jeng-Jung Wang†
Department of Information Engineering
I-Shou University, Kaohsiung, Taiwan 84008, R.O.C.

Tung-Yang Ho
Department of Industrial Engineering and Management
Ta Hwa Institute of Technology, Hsinchu, Taiwan 30740, R.O.C.

Ting-Yi Sung
Institute of Information Science
Academia Sinica, Taipei, Taiwan 115, R.O.C.

Abstract

A network is usually represented by a graph. The diameter of a graph is an important factor for communication as it determines the maximum communication delay between any pair of processors in a network. The diameter of a graph can be affected by adding or deleting edges. In this paper, we study the diameter variability arising from the change of edges in hypercubes.

1 Introduction

An interconnection network connects the processors of a parallel and distributed system. The topology of an interconnection network for a parallel and distributed system can always be represented by a graph, where each vertex represents a processor and each edge represents a vertex-to-vertex communication link. Communication is a critical issue in the design of a parallel and distributed system. The diameter of a graph is an important factor for communication as it determines the maximum communication delay between any pair of processors in a network.

Let $G = (V, E)$ be a graph. $V(G)$ is the vertex set of G and $E(G) \subseteq V(G) \times V(G)$ is the edge set of G. Let u and v be two different vertices in a graph G. The *distance* between u and v in G, denoted as $d_G(u, v)$, is the length of a shortest path joining them. The *diameter* of graph G, denoted as $D(G)$, is the maximum distance between any two vertices. The diameter of a graph determines the maximum communication delay between any pair of processors in a network [1] and can be affected by adding or deleting edges [2, 3, 4, 6, 7, 8, 9]. For example, let m-cycle C_m be a graph with vertex set $\{0, 1, 2, \ldots, m-1\}$ and edge set $\{(i, i+1) | 0 \le i \le m-1\}$, where addition is in integer modulo m. It is known that $D(C_m) = \lfloor m/2 \rfloor$. Let P_m be a graph with vertex set $\{0, 1, 2, \ldots, m-1\}$ and edge set $\{(i, i+1) | 0 \le i \le m-2\}$, it is known that $D(P_m) = m - 1$. It is easy to check that deleting any edge renders C_m to a path of m vertices and then the diameter increases to $m-1$. On the other hand, a cycle C_m can be obtained by adding the edge $(0, m)$ to path P_m. The diameter decreases to $\lfloor m/2 \rfloor$.

In the literature, there are several versions of the diameter variability arising from change of edges of a graph. For example, given a positive integer k and a graph G with diameter D, let $\Delta^{-k}(G)$ denote the maximum diameter of the graph obtained by deleting k edges from G, assuming that the resulting graph is still connected. For undirected graphs, some upper and lower bounds on $\Delta^{-k}(G)$ are summarized in the following table. For a survey of the known results related to the problem we refer the reader to [3].

	lower bound	upper bound	restrictions
Plesník [8]		$2D$	
Chung and Garey [4]	$(k+1)(D-3)$	$(k+1)D+k$	
Schoone, Bodlaender, and van Leeuween [9]	$(k+1)D-k$	$(k+1)D$	even D
	$(k+1)D-2k+2$	$(k+1)D$	odd $D \ge 3$
	$3D-1$	$3D-1$	$k=2$
	$4D-2$	$4D-2$	$k=3$
	$k+2$	$k+2$	$k \le 4$ and $k=6$
	$k+3$	$k+3$	$k=5$ and $k \ge 7$
Peyrat[7]	4	4	$D=2$
	$3\sqrt{2k}-3$	$3\sqrt{2k}+4$	$D=3$

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†Correspondence to: jjwang@isu.edu.tw.

A version of calculation of $\Delta^{-k}(G)$ is stated as follows. For given positive integers k, D and an undirected graph G, determine whether there exists a connected subgraph of G, obtained by deleting k edges from G, that has a diameter at least D [9]. It is quite difficult for general graphs, since it has been proved by Schoone et al. [9] to be NP-complete.

The other type of problem pertaining the change of diameter under the addition of edges is that given a graph G with diameter D, let $\Delta^{+k}(G)$ denote the minimum diameter of the graph obtained by adding k to G. For $k = 2$ or 3, Schoone et al. [9] showed that $\Delta^{+2}(P_n) = \lceil \frac{1}{3}n \rceil$ for $n \geq 4$, $\Delta^{+3}(P_n) = \lceil \frac{1}{4}(n+1) \rceil$ for $n \geq 5$, and $\Delta^{+2}(C_n) = \lceil \frac{1}{4}(n+2) \rceil$ for $n \geq 5$. For general $k \geq 3$ and $n \geq 5$, Chung and Garey [4] obtained the following results.

(1) $\frac{n}{k+1} - 1 \leq \Delta^{+k}(P_n) \leq \frac{n}{k+1} + 3$;

(2) $\frac{n}{k+2} - 1 \leq \Delta^{+k}(C_n) \leq \frac{n}{k+2} + 3$ if k is even;

(3) $\frac{n}{k+1} - 1 \leq \Delta^{+k}(C_n) \leq \frac{n}{k+1} + 3$ if k is odd.

Similarly, a version of calculation of $\Delta^{+k}(G)$ is defined by Schoone et al. [9] as follows. For given positive integers k, D and an undirected graph G, determine whether there exists supergraph of G, obtained by adding k edges to G, that has a diameter at most D. It is also quite difficult in general, since it has been proved by Schoone et al. [9] to be NP-complete.

In this paper, we consider the diameter variability arising from change of edges of graph G as follows. Let k be an arbitrary positive integer. We define

$D^{-k}(G)$**:** the least number of edges whose addition to G decreases the diameter by (at least) k;

$D^{+k}(G)$**:** the least number of edges whose deletion from G increases the diameter by (at least) k;

$D^{+0}(G)$**:** the maximum number of edges whose deletion from G does not change the diameter.

For example, $D^{+1}(C_m) = D^{+2}(C_m) = \cdots = D^{+(m-1-\lfloor m/2 \rfloor)}(C_m) = 1$ and $D^{-1}(P_m) = D^{-2}(P_m) = \cdots = D^{-(m-1-\lfloor n/2 \rfloor)}(P_m) = 1$. Note that $0 \leq D^{-i}(G) \leq D^{-j}(G)$ and $0 \leq D^{+i}(G) \leq D^{+j}(G)$ if $i \leq j$.

In this paper, we investigate how the change in the diameter of a hypercube graph with the addition of edges. An *n-dimensional cube*, Q_n, is a graph with 2^n vertices that can be one-to-one labeled with 0-1 binary strings so that two vertices are adjacent if and only if their labels differ in exactly one proposition. It is known that the diameter of Q_n is n. Graham and Harary [6] considered changing the diameter without considering the extent of the change, i.e., they considered $D^{-1}(G)$, $D^{+1}(G)$, and $D^{+0}(G)$. They showed that $D^{-1}(Q_n) = 2$ by adding two diagonal edges in any subcube Q_2 of Q_n, $D^{+1}(Q_n) = n-1$ by deleting $n-1$ edges in a subcube Q_{n-1} of Q_n which are incident at any specific vertex, and $D^{+0}(Q_n) \geq (n-3)2^{n-1} + 2$. Bouabdallah et al. [2] improved the lower bound of $D^{+0}(Q_n)$ and furthermore gave an upper bound, $(n-2)2^{n-1} - \binom{n}{\lfloor n/2 \rfloor} + 2 \leq D^{+0}(Q_n) \leq (n-2)2^{n-1} - \lceil (2^n-1)/(2n-1) \rceil + 1$.

In order to reduce the diameter of Q_n, we construct a supergraph of Q_n by adding edges to Q_n. The folded cube is an example of adding 2^{n-1} edges to Q_n such that diameter is reduced to $\lfloor n/2 \rfloor$, i.e., $D^{-\lfloor n/2 \rfloor}(Q_n) \leq 2^{n-1}$. In this paper, we employ the complementary edges to construct a supergraph $Q_{n;-k}$ of Q_n with diameter $n-k$. Then, we calculate the number of edges of $Q_{n;-k}$.

2 The Computation for $D^{-k}(Q_n)$

In this section, we construct a supergraph of Q_n, denoted by $Q_{n;-k}$, which will later be shown to reduce the diameter of Q_n by k. A $Q_{n;-k}$ is said to be optimal if it contains the least number of edges among all supergraphs of Q_n having the diameter $n-k$. For $k=1$, an optimal $Q_{n;-1} = Q_n \cup \{e_1, e_2\}$ is given by Graham and Harary [6], where e_1, e_2 are two diagonal edges in any 2-dimensional subcube Q_2 of Q_n [6].

Let $v = v_{n-1}v_{n-2}\ldots v_1v_0$ be a vertex in Q_n, where $v_i \in \{0,1\}$ for $0 \leq i \leq n-1$. Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two disjoint graphs, where each vertex in G_1 and G_2 is represented by an n-bit binary vector. Each vertex u in G_1 has the form $0u_{n-2}u_{n-3}\ldots u_1u_0$ and each vertex v in G_2 has the form $1v_{n-2}v_{n-3}\ldots v_1v_0$, where $u_i, v_i \in \{0,1\}$ for $0 \leq i \leq n-2$. We say that $G = G_1 \otimes G_2$ if $V(G) = V_1 \cup V_2$ and $E(G) = E_1 \cup E_2 \cup \{(u,v) \mid u \in V_1, v \in V_2$ and $(n-1)$-suffix of u equals to $(n-1)$-suffix of $v\}$. Then, we can write $Q_n = Q^0_{n-1} \otimes Q^1_{n-1}$, where each vertex in Q^i_{n-1} can be represented by $iv_{n-2}v_{n-3}\dots v_0$ with $v_{n-2}v_{n-3}\dots v_0 \in V(Q_{n-1})$ and $i = 0, 1$. Furthermore, Q^i_{n-1} is isomorphic to Q_{n-1}.

The n-dimensional folded cube, denoted by FQ_n, proposed in [5] is defined as

$$FQ_n = Q_n \cup \{(v, \bar{v}) \mid v \in V(Q_n)\}.$$

It is known that $D(FQ_n) = \lceil n/2 \rceil$ [5]. FQ_n is an example of adding 2^{n-1} edges to Q_n such that the diameter is reduced by $\lfloor n/2 \rfloor$.

Now, we construct $Q_{n;-k}$ for $n \geq 4$ and $2 \leq k \leq \lfloor n/2 \rfloor$ recursively as follows:

$$Q_{n;-k} = Q^0_{n-1;-k} \otimes Q^1_{n-1;-(k-1)} \tag{1}$$

with the initial conditions:

1. $Q_{2k;-k} = FQ_{2k}$ for $k \geq 2$,
2. $Q_{n;-1} = Q_n \cup \{e_1, e_2\}$,

where $Q^i_{n;-j}$ is isomorphic to $Q_{n;-j}$ with each vertex represented by an $(n+1)$-bit vector having i as the leftmost bit for $i = 0, 1$ and j positive integers.

Theorem 1 *$D(Q_{n;-k}) = n - k$ for $n \geq 4$ and $2 \leq k \leq \lfloor n/2 \rfloor$.*

Proof: We prove this lemma by induction on (n, k) in lexicographic ordering. Let $(n, k) = (4, 2)$. Then by the initial conditions of (1), we have $D(Q_{4;-2}) = D(FQ_4) = 2$. This completes the basis step.

Suppose that $D(Q_{m;-i}) = m - i$ whenever (m, i) is less than (n, k) in lexicographic ordering. Because $Q^0_{n-1;-k}$ is isomorphic to $Q_{n-1;-k}$ and $(n-1, k)$ is smaller that (n, k), the hypothesis tells us that $D(Q_{n-1;-k}) = n-k-1$. Similarly, we have $D(Q_{n-1;-(k-1)}) = n-1-(k-1) = n-k$. By (1), we have $D(Q_{n;-k}) \geq n - k$. Thus, it suffices to show that $d_{Q_{n;-k}}(0u, 1v) \leq n - k$ for $0u$ is any vertex in $Q^0_{n-1;-k}$ and $1v$ is any vertex in $Q^1_{n-1;-(k-1)}$. Any shortest path from $0u$ to $1v$ in $Q_{n;-k}$ is given by appending the vertex $1v$ to the shortest path from $0u$ to $0v$. Since $d_{Q^0_{n-1;-k}}(0u, 0v) \leq n - k - 1$, it follows that $d_{Q_{n;-k}}(0u, 1v) \leq n-k$. This finishes the inductive step. Hence, the theorem follows.

Now, we calculate how many number of edges added to Q_n so that the diameter of Q_n decreases to $n - k$. Let $\Delta_{n,k} = |E(Q_{n;-k})| - |E(Q_n)|$. It is trivially $\Delta_{n,1} = 2$ and $\Delta_{2k,k} = 2^{2k-1}$. By (1), we have

$$\begin{aligned}
\Delta_{n,k} &= |E(Q_{n;-k})| - |E(Q_n)| \\
&= |E(Q^0_{n-1;-k} \otimes Q^1_{n-1;-(k-1)})| - |E(Q_n)| \\
&= |E(Q^0_{n-1;-k})| + |E(Q^1_{n-1;-(k-1)})| \\
&\qquad +2^{n-1} - |E(Q_n)| \\
&= \Delta_{n-1;-k} + \Delta_{n-1;-k+1} + |E(Q_{n-1})| \\
&\qquad +|E(Q_{n-1})| + 2^{n-1} - |E(Q_n)| \\
&= \Delta_{n-1;-k} + \Delta_{n-1;-k+1}.
\end{aligned} \tag{2}$$

In addition, by recursively applying (1), $\Delta_{n,k}$ can be described by the recurrence

$$\Delta_{n,k} = 2^{2k-1} + \sum_{i=2k}^{n-1} \Delta_{i,k-1} \qquad \text{for } n \geq 2k. \tag{3}$$

For $n \geq 2k$ and $k \geq 2$, we claim a solution to (3) to be

$$\begin{aligned}
\Delta_{n,k} &= 2^k + 2^{k-1}\binom{n-k}{1} \\
&\quad +2^{k-2}\binom{n-(k-1)}{2} \\
&\quad +\dots + 2\binom{n-2}{k-1}.
\end{aligned} \tag{4}$$

We prove (4) by induction on (n, k) in lexicographic ordering. Let $(n, k) = (4, 2)$. Then $\Delta_{4,2} = 2^3 = 8 = 2^2 + 2\binom{2}{1}$. Let $(n, k) = (5, 2)$. Then $\Delta_{5,2} = 2^3 + 2 = 10 = 2^2 + 2\binom{3}{1}$. This completes the basis step.

Suppose that $\Delta_{m,i} = 2^i + 2^{i-1}\binom{m-i}{1} + 2^{i-2}\binom{m-(i-1)}{2} + \dots + 2\binom{m-2}{i-1}$ whenever (m, i) is less than (n, k) in lexicographic ordering. By (2) and the hypothesis, we have

$$\begin{aligned}
\Delta_{n,k} &= \Delta_{n-1;-k} + \Delta_{n-1;-(k-1)} \\
&= 2^k + 2^{k-1}\binom{n-1-k}{1} \\
&\quad +2^{k-2}\binom{n-1-(k-1)}{2}
\end{aligned}$$

$$
\begin{aligned}
&+\ldots+2\begin{pmatrix} n-3 \\ k-1 \end{pmatrix}+2^{k-1} \\
&+2^{k-2}\begin{pmatrix} n-1-(k-1) \\ 1 \end{pmatrix} \\
&+2^{k-3}\begin{pmatrix} n-1-(k-2) \\ 2 \end{pmatrix} \\
&+\ldots+2\begin{pmatrix} n-3 \\ k-2 \end{pmatrix} \\
&=2^k+2^{k-1}+2^{k-1}\begin{pmatrix} n-1-k \\ 1 \end{pmatrix} \\
&+2^{k-2}\begin{pmatrix} n-(k-1) \\ 2 \end{pmatrix}+\ldots+2\begin{pmatrix} n-2 \\ k-1 \end{pmatrix} \\
&=2^k+2^{k-1}\begin{pmatrix} n-k \\ 1 \end{pmatrix} \\
&+2^{k-2}\begin{pmatrix} n-(k-1) \\ 2 \end{pmatrix}+\ldots+2\begin{pmatrix} n-2 \\ k-1 \end{pmatrix}.
\end{aligned}
$$

This completes the inductive step. Thus, the following theorem can be immediately obtained.

Theorem 2 $D^{-k}(Q_n) \leq 2^k + 2^{k-1}\begin{pmatrix} n-k \\ 1 \end{pmatrix} + 2^{k-2}\begin{pmatrix} n-(k-1) \\ 2 \end{pmatrix} + \ldots + 2\begin{pmatrix} n-2 \\ k-1 \end{pmatrix}$.

3 Conclusion

In this paper, we have studied the diameter variability arising from the change of edges for hypercubes. In particular, we construct supergraphs of these given graphs such that the diameter is reduced by a constant k. An area for future researches are the exact values of $D^{-k}(Q_n)$, the relationship between $D^x(G_1 \times G_2)$ and $D^x(G_1)$ and $D^x(G_2)$, where $G_1 \times G_2$ is the Cartesian product of G_1 and G_2 and x is one of $-k$, $+k$, or $+0$.

References

[1] J.-C. Bermond and B. Bollobas, The diameter of graphs – A survey, *Congr. Numer.* 32 (1981) 3–27.

[2] A. Bouabdallah, C. Delorme, and S. Djelloul, Edge deletion preserving the diameter of the hypercube, *Discrete Applied Mathematics*, 63, (1995), 91–95.

[3] F. R. K. Chung, Diameters of communication networks, In *Mathematics of Information Processing, Proceedings of Symposia in Applied Mathematics*, 34 (M. Anshel and W. Gewirtz, eds.), *American Mathematics Society, Providence*, RI (1986) 1–18.

[4] F. R. K. Chung and M. R. Garey, Diameter bounds for altered graphs, *Journal of Graph Theory* 8(4) (1984) 511–534.

[5] A. El-Amawy and S. Latifi, Properties and performance of folded hypercubes, *IEEE Trans. Parallel and Distributed Systems*, 2, (1991), 31–42.

[6] N. Graham and F. Harary, Changing and unchanging the diameter of a hypercube, *Discrete Applied Mathematics*, 37/38, (1992), 265–274.

[7] C. Peyrat, Diameter vulnerability of graphs, *Discrete Applied Mathematics* 9 (1984) 245–250.

[8] J. Plesník, Note on diametrically critical graphs, *In Recent Advances in Graph Theory* (*Proceedings of the 2nd Czechoslovak Symposium, Prague*, (1974)), *Prague: Academia* (1975) 455–465.

[9] A. A. Schoone, H. L. Bodlaeder, and J. van Leeuween, Diameter increase caused by edge deletion, *Journal of Graph Theory*, 11(3) (1987) 409–427.